HUX / V1-PROOF / ELECTRICAL

Complete system wiring by location

Rear elevation of the R05 body. Robot LEFT (near wall) is at the left. Each board sits where it mounts. Ports are connector positions. Every harness conductor is drawn.

SOURCE-CHECKED PLAN
HARNESS / BENCH RELEASE OPEN

LEFT LEG · NEAR WALL SIDE
fixture buttons, pack lead, left gearmotor

READING THE SHEET
Wire IDs → searchable conductor register.
Bundle cut lengths → EL-06 and the harness register.
Port labels: connector position / pin.
Dots connect. Gaps at crossings do not.
Dashed boards need part selection and bench checks.
BLACK wheel lead is switched OUT2, not ground.
XH = JST XH 2.50 mm. XT30 = 12 V power path.
1x6 0.1 in = the Pololu pigtail housing.

CTRL DECK · Z +170 · near half: IMU · far half: Pico 2

U1 / SparkFun LSM6DSO
3.3 V · main SPI pads · open the ADDR jumper for SPI · EL-02

JMAIN.5 / SDO — W002 — PH-C.1 / GP12
JMAIN.6 / CS — W003 — PH-C.2 / GP13
JMAIN.1 / GND — W004 — PH-C.3 / GND
JMAIN.4 / SCL — W005 — PH-C.4 / GP14
JMAIN.3 / SDA / SDI — W006 — PH-C.5 / GP15
JINT.2 / INT1 — PH-C.6 / GP16
JINT.1 / INT2
JMAIN.2 / 3V3 — W008 — PH-F.1 / 3V3(OUT)

SP1 splice

JMAIN: B6B-XH-A on the six main pads.
JINT: B2B-XH-A on INT2 / INT1.
INT2 stays empty.
Ports are in cable order; the position number gives the pad.

P1 / Pico 2
RP2350 · 3.3 V GPIO · eight XH headers in contiguous pin groups · GP25 LED onboard

PH-F.2 / 3V3_EN
PH-F.3 / GND
PH-F.4 / VSYS
PH-G.1 / GP0
PH-G.2 / GP1
PH-G.3 / GND

Headers: PH-A p4–p7, PH-A2 p8–p10, PH-B p11–p15, PH-C p16–p21, PH-D p24–p29, PH-E p31–p34, PH-F p36–p39, PH-G p1–p3.
Desolder the pre-soldered pins, then solder the XH headers to the board (2026-10-17).
NC: ADC_VREF p35, RUN p30, 3V3_EN p37, GP17 p22, GP19–GP21.
EL-02 shows the physical board and the IMU pins.

P1 / Pico 2

USB.1 / VBUS — W041 — CABLE.1 / VBUS
USB.2 / D+ — W042 — CABLE.2 / D+
USB.3 / D- — W043 — CABLE.3 / D-
USB.4 / GND — W044 — CABLE.4 / GND

RIGHT HARNESS TRUNK
down the front face of the body

PH-A.1 / GP2
PH-A.2 / GP3
PH-A.3 / GP4
PH-A.4 / GP5
PH-A2.1 / GND
PH-A2.2 / GP6 — W049
PH-A2.3 / GP7 — W050
PH-B.1 / GP8 — W054
PH-B.2 / GP9 — W055
PH-B.3 / GND — W053
PH-B.4 / GP10
PH-B.5 / GP11 — W051
PH-D.1 / GP18 — W056
PH-D.2 / GP19
PH-D.3 / GP20
PH-D.4 / GP21
PH-D.5 / GND — W058
PH-D.6 / GP22
PH-E.1 / GP26
PH-E.2 / GP27
PH-E.3 / AGND
PH-E.4 / GP28

RIGHT LEG · FAR WALL
USB cable, right gearmotor

JUSB / Laptop USB
CABLE.1 / VBUS
CABLE.2 / D+
CABLE.3 / D-
CABLE.4 / GND

Micro-USB cable to the front of the Pico 2.

NEAR WALL · LEFT

FAR WALL · RIGHT

BUS DECK · Z +120 · near half: power distribution · far half: spare cassette position

JPICO.1 / VSYS — JPICO.2 / GND

PDB / Power distribution

From the unswitched fused bus. D5 Schottky → VSYS.
Logic stays on through a kill. 5 V to JLEG.
5 V regulator · part TBD

J5V.1 / 5V
J5V.2 / GND
JIN.1 / BUS
JIN.2 / GND

SPARE CASSETTE POSITION
Was the servo bus and the encoder conversion. No board in V1-PROOF.
The encoder dividers sit on JLEG.

W011
W010
W046
W047
W048
W045
W009

PWR DECK · Z +60 · near half: kill chain · far half: driver board

JBUS.1 / BUS
JBUS.2 / GND
JCTL.1 / nSLEEP
JCTL.2 / spare
J3V3.1 / 3V3
J3V3.2 / spare

KC / Kill chain

JKILL.1 / 3V3K
JKILL.2 / OFF node
JARM.1 / 3V3K
JARM.2 / ON node
XPACK.1 / +
XPACK.2 / -

Pololu 2813 + 74HC14 · EL-04
XPACK: fused pack in.
XRAIL: 2813 VOUT = the switched rail.
LD1117V33 gate logic from the bus.
100k / 22k rail divider → RAIL_ADC.
KILL → OFF. ARM → ON via 4.7k, clamped by HB_OK.
Not ordered or tested. K0–K6 open.

JHB.1 / HB
JHB.2 / RAIL_ADC
JHB.3 / AGND
XRAIL.1 / +
XRAIL.2 / -

DB / Driver board
DL, DR: Pololu 4035 / DRV8874 carriers on one 52 x 38 mm board · EL-03, EL-05

10 kΩ nFAULT pull-ups to 3V3. 1 kΩ CS series.
VREF divider ≈ 2.8 V (≈ 2.5 A). PMODE, IMODE to GND. VM NC.
nSLEEP from GP10 to both carriers.
EN low = brake. nSLEEP low = outputs off.
OUT1 / OUT2 go to JLEG, not to the motor directly.

W014
W015
XRAIL.1 / +
XRAIL.2 / -

JSIG-L.1 / EN
JSIG-L.2 / PH
JSIG-L.3 / nFAULT
JSIG-L.4 / CS out
JSIG-L.5 / GND
JSIG-R.1 / EN
JSIG-R.2 / PH
JSIG-R.3 / nFAULT
JSIG-R.4 / CS out
JSIG-R.5 / GND

SWK / KILL
LEADS.1 / NO — W037 — JKILL.1 / 3V3K
LEADS.2 / COM — W038 — JKILL.2 / OFF node
red mushroom · momentary NO · fixture

SWA / ARM
LEADS.1 / NO — W039 — JARM.1 / 3V3K
LEADS.2 / COM — W040 — JARM.2 / ON node
green · momentary NO · fixture

BAT / 3S pack
LEADS.1 / + — FUSE.1 — W035 — XPACK.1 / +
LEADS.2 / - — FUSE.2 — XPACK.2 / -
on the tray · XT30 · TBD
inline holder · value TBD

The pack sits on the tray at the bottom of the body. It is drawn here so that its lead to the kill chain is clear.
Both buttons are on the bench fixture, not on the robot.

JML.1 / OUT1
JML.2 / OUT2
JMR.1 / OUT1
JMR.2 / OUT2

W060
W061
W062
W063

UNDER THE DECKS · leg junction board under the PWR deck · 3S pack on the tray at Z −35

W016
J5V.1 / 5V
J5V.2 / GND
JPL.1 / OUT1
JPL.2 / OUT2
JPR.1 / OUT1
JPR.2 / OUT2
JENC.1
JENC.2
JENC.3
JENC.4
JENC.5

ML / Pololu 4752 · LEFT WHEEL

ML
12 V · 30:1 · encoder

PIGTAIL.1 / RED — W022 — JML_IN.1 / RED
PIGTAIL.2 / BLACK — W023 — JML_IN.2 / BLACK
PIGTAIL.3 / GREEN — W024 — JML_IN.3 / GREEN
PIGTAIL.4 / BLUE — W025 — JML_IN.4 / BLUE
PIGTAIL.5 / YELLOW — W026 — JML_IN.5 / YELLOW
PIGTAIL.6 / WHITE — W027 — JML_IN.6 / WHITE

JLEG / Leg junction

Accepted 2026-10-17 · under the PWR deck · 1x6 pigtail headers · 2.2 kΩ / 3.3 kΩ dividers
Motor leads pass to JPL / JPR. Encoder A / B 5 V → 3.0 V → JENC.
Board layout, mount detail and the service loop for later leg motion remain open.

JMR_IN.1 / RED — W028 — PIGTAIL.1 / RED
JMR_IN.2 / BLACK — W029 — PIGTAIL.2 / BLACK
JMR_IN.3 / GREEN — W030 — PIGTAIL.3 / GREEN
JMR_IN.4 / BLUE — W031 — PIGTAIL.4 / BLUE
JMR_IN.5 / YELLOW — W032 — PIGTAIL.5 / YELLOW
JMR_IN.6 / WHITE — W033 — PIGTAIL.6 / WHITE

MR / Pololu 4752 · RIGHT WHEEL

MR
12 V · 30:1 · encoder